

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and built **AI-native interfaces** for developer workflows, integrating **LLM tooling** and **agent architectures** to improve platform usability.
- Developed and maintained **evaluation frameworks** and **instrumentation systems** to measure the reliability and performance of AI agents, reducing task failure rates by **18%**.
- Engineered **reusable patterns and infrastructure** for AI integration across applications, including **Edge Functions** and **embeddings**, centralizing shared logic across multiple services.
- Collaborated with **documentation and frontend teams** to expose product knowledge effectively to AI agents, improving the accuracy of AI-generated responses by **22%**.
- Implemented **Postgres-backed vector databases** with **pgvector** for efficient similarity search on large datasets, enabling faster retrieval for AI agent context.
- Authored production runbooks and internal documentation that streamlined the onboarding of new internal tooling features, reducing initial user friction by an estimated **30%**.

SKILLS

Languages – JavaScript, TypeScript, Python, SQL, Java, R

AI & ML – LLM Tooling, Agent Architectures, Embeddings, Vector Databases, pgvector, AI Evaluation Frameworks, scikit-learn, R

Backend & Infrastructure – Node.js, NestJS, TypeScript, REST APIs, Spring Boot, Edge Functions, Docker, Kubernetes, PostgreSQL, npm, ES modules, bundlers

Developer Tooling – Developer Tools, Internal Systems, CI/CD, Jenkins, Git, API Development, CLI

Data & Analytics – Data Modelling, Data Warehousing, Power BI, Tableau, DAX, Power Query, NumPy, pandas, Statistical Analysis

PROJECTS

AI Agent Skill Development Platform

- Designed and built a platform for defining and managing **agent skills**, enabling developers to integrate custom logic with **LLM capabilities**.
- Implemented a **TypeScript**-based workflow engine using **Node.js** that orchestrated calls to external APIs and internal services based on agent directives.
- Integrated **PostgreSQL** with **pgvector** to store and query embeddings for knowledge retrieval, improving the context-awareness of AI agents by **25%**.

Developer Dashboard AI Assistant

- Developed an **AI-driven dashboard assistant** to support debugging and self-serve product usage, translating complex system logs into actionable insights.
- Built **Python scripts** to preprocess and structure data for training the assistant's underlying models, handling approximately **5,000 log entries per hour**.
- Designed and implemented a feedback loop mechanism, allowing developers to rate assistant responses, which informed iterative improvements and reduced incorrect suggestions by **15%**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)